A conditional TNF-α binder, and the verifiers that gated it

Adaptyv × Anthropic Protein Design Competition, Challenge 02

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Abstract

We designed candidate binders to human TNF-α required to satisfy three objectives at once: bind human TNF-α, cross-react with the mouse ortholog, and bind at pH 7.4 while releasing at pH 6.0. Four generative pipelines proved substantially interchangeable; the outcome was determined almost entirely by *verification*. Of 57 constructs entering the gate stack, 3 were submitted and 2 passed the organiser's novelty check — a yield of $2/57 \approx 3.5\%$. The highest-scoring construct by every interface metric was verbatim human ubiquitin. One design was submitted and failed the organiser's automatic novelty check, establishing empirically that *fold* similarity, not sequence novelty, is the binding constraint. We report the full funnel including failures, and state explicitly what remains unmeasured: there is no measured binding of any kind.

1 Target and epitope

The epitope was computed rather than taken from literature. TNF-α residues within 4.5 of TNFR2 in PDB `3ALQ` give 33 contact positions. Intersecting these with a human/mouse alignment of the soluble domain ($124/157 = 79.0\%$ identity, reproducing the organiser's stated figure) partitions them into 24 conserved and 9 divergent positions: P20H, R31Q, S71D, T72–, H73Y, V85I, T89E, I97V, A111L.

R31→Q is the trap. It carries the largest solvent-accessible surface area in the entire epitope ($193\,^2$) and sits adjacent to R32 in the 29–33 loop engaged by adalimumab and infliximab. This is why neither therapeutic is mouse cross-reactive. Any design gripping that loop satisfies Objective 1 and forfeits Objective 2.

Two conserved sub-patches were considered: a lobe (S86, Y87, Q88, K90, V91, N92) at 90% conservation, and a pocket (113, 115, 143, 144, 146, 147) at 100%. A screen found no significant difference between them (Table 1).

2 Generators and the levers that did not work

Table 1: Design levers, measured as binder-to-target ipTM. Only target representation mattered.

Lever	Contrast	Δ ipTM
Target representation	A+B dimer → A+B+C trimer	+0.11
Architecture	de novo vs. nanobody graft	0.00
Topology conditioning	free vs. 3-helix bundle	−0.03
Epitope	conserved lobe vs. conserved pocket	0.00

Truncating the physiological trimer to a dimer — a choice made to save GPU memory — cost 0.11 in interface confidence, more than any deliberate design decision contributed.

3 The verifier stack

Each verifier below caught something that had already passed every check preceding it.

3.1 Composition

A model-free check on amino-acid usage. It eliminated 50 of 57 constructs. The necessity is established by a single observation: a sequence that is 59% alanine received a binder-to-target ipTM of 0.556 and ranked first in a 40-design shortlist. Interface confidence models score degenerate sequences confidently.

3.2 Sequence novelty

Exact 20-mer matching against PDB `seqres` (1.17M chains) and Swiss-Prot (576k). A genuine de novo sequence essentially never shares a 20-mer with a natural protein, making this decisive rather than statistical. It rejected the single highest-scoring construct of the campaign — best interface score, lowest interface PDE (0.89), an ideal-pKa histidine — which was **verbatim human ubiquitin**.

3.3 Interface

All interface quantities derive from per-residue buried surface area (unbound minus bound SASA). Contact-pair counting was implemented first and was actively misleading: counts scale with interpenetration, so clashing poly-alanine designs out-scored real ones on every ratio built from them.

3.4 Fold novelty

The organiser scores novelty 1–4 and requires ≥ 3; level 4 demands sequence identity $\leq 30\%$ *and* less than moderate structural similarity, where similarity is TM-score weighted by coverage. We implemented this with FoldSeek and TM-align over CATH50.

Empirical calibration. One submitted design **failed** the organiser's automatic check and two passed. The three differed by 0.6 percentage points in sequence identity (37.5% vs. 38.1%), so sequence novelty cannot explain the split. The failing design matched CATH 2.60.40 — the immunoglobulin-like β-sandwich — at TM-score 0.893 over 88.8% coverage. Fold similarity decided it.

Our gate predicts the failure correctly but *under-scores* both passes, returning 2 where the organiser returned ≥ 3. It is therefore a conservative pre-filter: decisive on failure, advisory on borderline. An earlier claim that it reproduced all three verdicts was incorrect, arising from a missing sequence-identity value defaulting to zero.

3.5 pH switch

Objective 3 requires a histidine whose pK_a lies *between* the two assay pH values: neutral at 7.4 so the binder binds, cationic at 6.0 so it releases. Three mechanisms were derived from first principles and each was refuted by PROPKA predictions on real designs (Table 2).

No structural descriptor predicted switchability. Positions therefore come from an exhaustive per-position scan: 200 substitutions yielded 18 ideal (pK_a 6.3–7.2) and 17 usable (6.0–7.6). Multi-histidine combinations were verified to retain their pK_a within 0.21 units of the single-mutant values, supporting a cooperative switch.

Table 2: Proposed pK_a mechanisms against measurement ($n = 50$ substitutions).

Hypothesis	Prediction	Measured
Buried His pays desolvation penalty	raises effective switch	$r = -0.37$; buried sites at pK_a 3.1–3.6
His adjacent to cation is repelled	lowers pK_a usefully	$r = +0.11$ (wrong sign)
Carboxylate proximity raises pK_a	raises into window	$r = +0.21$; net $+6 \rightarrow$ pK_a 3.6

4 Cross-model disagreement

No single confidence score is treated as truth, because the models disagree on identical structures (Table 3).

Table 3: Binder-to-target interface confidence by predictor.

Structure	AlphaFold2	Boltz-2	BoltzGen folding
BindCraft (AF2-optimised) design	**0.92**	**0.21**	—
BoltzGen designs	—	0.44–0.64	~ 0.20

AlphaFold2 rating its own optimised designs at 0.92 where an independent predictor returns 0.21 is self-confirmation, not validation. Two further artefacts were identified and corrected: *global* ipTM is unusable on a multi-chain target (the TNF trimer's internal interfaces score ~ 0.88 and dominate the mean, so a dimer→trimer change lifted global ipTM $0.20 \rightarrow 0.52$ while the binder interface stayed flat), and best-of-n reporting inflates (metrics are stored per diffusion sample; we rank on the mean).

Measured across all 57 folded constructs, global ipTM overstates the designed interface by $+0.405$ on average — roughly half its own value. Every interface number in this report is instead a `pair_chains_iptm` entry between the binder chain and each target chain, symmetrised over both matrix directions (which differ by up to 0.08), averaged across target chains, then averaged across diffusion samples. Best-of-n and mean differ by up to 0.35 within a single construct.

4.1 Confidence was not sufficiency

Six constructs cleared the ~ 0.50 threshold conventionally taken to indicate a credible de novo interface. *None was submittable*, and the reasons were not subtle (Table 4).

Table 4: The six highest-confidence constructs, and why each was rejected.

Construct	ipTM	Verdict
`pocket_free_7__A63H`	0.642	37% glycine; 3 steric clashes
`pocket_free_1__A73H`	0.634	clean sequence; 18% of interface on epitope
`pocket_free_2__A76H`	0.617	clean sequence; 13% of interface on epitope
`pocket_helix_1__A20H`	0.583	59% alanine; 3 steric clashes
`pocket_helix_1`	0.556	59% alanine; 4 steric clashes
`pocket_helix_1__A52H`	0.526	59% alanine; 3 steric clashes

The second and third rows are the instructive ones: real sequences, genuine interface confidence, and an interface placed somewhere other than the conserved patch they were conditioned on. Epitope focus — not confidence — was the operative constraint, and a gate reading confidence alone would have passed all six. This is also why the remaining generator is BindCraft2 with forced targeting and hard coldspot filters rather than further BoltzGen sampling.

5 Funnel

Table 5: Full funnel, failed generations included in the denominator.

Stage	n	Attrition
Constructs entering gates	57	—
Passed sequence novelty	53	4 memorised (2 ubiquitin, 2 VHH framework)
Passed composition	7	46 lost to Ala/Gly degeneracy
Passed interface quality	3	clash, tiny interface, off-epitope
Submitted	3	—
Passed organiser novelty	**2**	1 failed on fold similarity

Yield from generation to novelty-passing: $2/57 \approx 3.5\%$.

Declared deviation. These filters were not all fixed before observing results. The composition gate was added after poly-alanine appeared at the top of a shortlist, and the fold gate after a submitted design failed. The funnel above is therefore not a pre-registered statistic.

6 Reproducibility

The gate stack is installed and run as one command-line tool; each subcommand prints the measurement a claim in this report rests on.

```
tnf-binder epitope                    # the computed epitope and its divergent positions
tnf-binder confidence <predictions/>  # rank designs on mean binder-to-target ipTM
tnf-binder verify <complex.cif>       # composition / interface / pH verdicts for one design
tnf-binder validate <submission/>     # submission format pre-flight
tnf-binder wandb-log <assets/>        # funnel, tables and structures to Weights & Biases
```

The campaign run, with all 57 constructs' confidence tables, the pK_a scans and the folded complexes as interactive 3D media, is at wandb.ai/chaleong/tnf-conditional-binder. A report built from that run's own panels — every table and every folded complex embedded rather than described — is here, and an interactive walkthrough of the architecture is published as a Claude Artifact.

7 What is not established

- **No measured binding of any kind.** No expression, size-exclusion chromatography, thermal stability, or SPR/BLI. Every number here is computational.
- pH selectivity rests on predicted pK_a. No folding model represents protonation state, and the assay is not simulated.
- Mouse cross-reactivity rests on a *predicted* mouse structure and conserved-epitope coverage, not on measured binding to mouse TNF-α.
- Interface confidence is model-dependent and unresolved; we report a range.
- No validated mechanism for the pH switch replaces the three that were refuted. Positions are empirical.

A generated sequence that has passed computational checks is a candidate. A result is a measured biological endpoint on a preserved funnel, with controls. This work produced the former.